

Zhuoli Ouyang | Curriculum Vitae

Beijing – China

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Personal Information

Date of Birth: 2004-10-15

Nationality: China

GPA: 3.94/4.00 (last year), 3.88/4.00 (total)

Languages: English, Chinese, Japanese

Research Interest

I am an undergraduate student with a strong foundation in mathematics and AI. My research interests span **generative models**, **reinforcement learning**, and **optimization**, with a focus on building more **efficient AI** systems. I am particularly excited about **embodied AI** and the challenge of enabling agents to perceive, reason, and act in complex environments. I aim to bridge principled mathematical thinking with practical system design, and look forward to contributing to research at the frontier of these areas.

Education

Southern University of Science and Technology—Second-year undergraduate

Information Engineering, Dept. of Electronic & Electrical Engineering, China

GPA: 3.88, Rank: **4/46**, Honor Class

Signals and Systems (97/100) [Signal & Systems](#), Engineering Mathematics (95/100), Probability Theory and Mathematical Statistics (95/100)

Shenzhen

2024–Present

Southern University of Science and Technology—First-year undergraduate

Department of Mathematics, China

GPA: 3.86, Rank: **94/1346** (overall in university), **Top 7%**, Fields Elite Class (Honor Class)

Algebraic Geometry (Graduate level), Advanced Functional Analysis (Graduate level), Analysis (97/100, Honor) [Analysis](#), Advanced Algebra (97/100, Honor), C programming (95/100)

Shenzhen

2023–2024

Tsinghua University High School

High School Diploma, China

Beijing

2020–2023

School Affiliated to Tsinghua University

Secondary Education, China

Beijing

2017–2020

Honors & Awards

2026: ALT 2026 Best Student Paper Award

2026: ISIT 2026 Strong Accept

2026: DATE 2026 Oral Paper (ACEMARL)

2024: SUSTech, Department of Electronics and Electrical Engineering-(Honor Class)

2023: SUSTech, Mathematics Department - Fields Elite Class (Top 1%, Honor Class)

Publications

RMNP: Row-Momentum Normalized Preconditioning for Scalable Matrix-Based Optimization *ICML 2026*
Shenyang Deng, Zhuoli Ouyang* et al. (*Equal contribution)* [\[arXiv\]](#) [\[ICML\]](#) [\[Project Page\]](#)

Understanding Generalization from Embedding Dimension and Distributional Convergence *ICML 2026*
Junjie Yu, Zhuoli Ouyang*, Haotian Deng*, Chen Wei, Wenxiao Ma et al. (*Equal contribution)* [\[arXiv\]](#) [\[ICML\]](#)

Depth, Not Data: An Analysis of Hessian Spectral Bifurcation *ISIT 2026*
Shenyang Deng, Boyao Liao*, Zhuoli Ouyang* et al. (*Equal contribution)* [\[arXiv\]](#) [\[Project Page\]](#)

Suspicious Alignment of SGD: A Fine-Grained Step Size Condition Analysis *ALT 2026*
Shenyang Deng, Boyao Liao, Zhuoli Ouyang et al. [\[arXiv\]](#) [\[Project Page\]](#)

Doloris: Dual Conditional Diffusion Implicit Bridges with Sparsity Masking Strategy for Unpaired Single-Cell Perturbation Estimation *ICLR 2026*
Changxi Chi, Jun Xia, Yufei Huang, Zhuoli Ouyang et al. [\[OpenReview\]](#)

ACEMARL: Adaptive Clustering Enhanced Multi-Agent Reinforcement Learning for Analog Circuit Sizing *DATE 2026*

Han Wu*, Haoning Jiang*, Zhuoli Ouyang et al. (* Equal contribution)

[\[ResearchGate\]](#)

FD-MAGRPO: Functionality-Driven Multi-Agent Group Relative Policy Optimization for Analog-LDO Sizing *AAAI 2026*

Haoning Jiang*, Han Wu*, Zhuoli Ouyang et al. (* Equal contribution)

[\[AAAI\]](#)

Research & Study Experience

Dartmouth College

Hanover, NH

Research Intern – Optimizers & Learning Theory

2025–Present

Advisor: [Yaoqing Yang](#)

Working on optimization algorithms and learning theory, contributing to theoretical analysis and algorithm design for machine learning.

Achievement: ICML 2026; ALT 2026 (Best Student Paper Award); ISIT 2026

Shanghai AI Laboratory / OpenDataLab

Shanghai

Research Intern – Optimizer Benchmark & Survey

2025–Present

Advisors: [Cheng Tan](#), [Siyuan Li](#)

Conducting a comprehensive benchmark and survey of optimization algorithms for large-scale AI, covering classical and modern optimizers across convergence, efficiency, and generalization.

AI for Scientific Simulation and Discovery Lab, Westlake University

Hangzhou

Research Intern

2025–Present

Advisor: [Tailing Wu](#)

Working on generative AI methods (diffusion models, flow matching, stochastic optimization) for scientific simulation and discovery. Research spans AI for physical sciences—including fluid dynamics, plasma control, and embodied intelligence—as well as AI for life science, focusing on generative modeling of biological systems and evolutionary mechanisms.

HKUST (Guangzhou)

Guangzhou

Research Intern – Theory-First Learning & Optimization for AI

2025–Present

Advisor: [Yao Shu](#)

Working on foundational optimization and learning theory for large-scale AI systems. Research spans black-box optimization (zeroth-order, Bayesian, bandit, RL), learning dynamics (generalization, stability, robustness), and their application to LLMs and intelligent agents.

Neural Computing and Control Lab (NCC Lab), SUSTech

Shenzhen

Research Intern – Generalization Theory for Deep Learning

2025–Present

Advisor: [Quanying Liu](#)

Working on generalization theory for deep learning, understanding generalization from embedding dimension and distributional convergence.

Achievement: ICML 2026

SUSTech

Shenzhen

AI for EDA

2025

Advisor: [Junmin Jiang](#)

Applying Reinforcement Learning and AI algorithms to analog circuit design.

Achievement: AAAI 2026, DATE 2026, TCAD

Other Skills

Programming: Python, C, C++, MATLAB, LaTeX, etc.

Software: PyTorch, PETSc, MATLAB, Fluent

Soccer: Second-class National Athlete in Football

Piano: Level 8, Central Conservatory of Music, Beijing

Hobbies: Exploring mathematics, Coding, Philosophy and History, Playing piano, Playing soccer